

Tewodros Abisso

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SUMMARY

Third-year Software Engineering student at Seneca Polytechnic seeking a summer software engineering internship. Experience building full-stack, backend, and AI-powered applications with Python, JavaScript, and Java, and designing accessibility-focused UI as a UI Designer at Signai. Skilled in REST API development, cloud deployment (AWS, Docker), database-driven systems, and software engineering practices including debugging, testing, and Agile development.

EDUCATION

Bachelor of Software Engineering (B.Eng.), Seneca Polytechnic, Toronto, ON *Expected Jun 2028*

Relevant Coursework: Operating Systems | Data Structures & Algorithms | Artificial Intelligence | Software Testing & QA | Software Design & Deployment | Databases

EXPERIENCE

UI Designer, Signai *May 2026 – Aug 2026*

- Designed the user interface for a camera-based sign language translation app that helps deaf and hard-of-hearing users communicate by signing to their device.
- Wrote user stories and mapped user flows to define core app features before design began.
- Implemented UI designs in Figma with a focus on accessibility and ease of use for users with hearing loss.

PROJECTS

Business AI Agent Platform | [GitHub](#)

Python | FastAPI | LangChain | RAG | AWS | Docker | Render

- Designed and implemented a modular AI automation platform for task planning, workflow orchestration, and content generation.
- Developed RESTful APIs with FastAPI to expose AI services and support a scalable backend architecture.
- Built data ingestion pipelines and retrieval-augmented generation (RAG) workflows using vector embeddings and document indexing to enable contextual AI responses.
- Implemented unit testing, validation checks, logging, and structured debugging to improve API reliability and stability.
- Containerized backend services with Docker and deployed applications on AWS EC2.
- Deployed the backend on Render with environment-based configuration for production hosting.

Smart Garage Service Management System

JavaScript | Node.js | SQL | AWS EC2

- Developed a full-stack service management system to handle booking, employee management, and service workflows.
- Implemented REST APIs and role-based authentication for secure access control.
- Designed relational database schemas to manage operational data and customer records.
- Performed debugging, log analysis, and structured testing to ensure stable backend services.
- Deployed application services to AWS EC2 cloud infrastructure.

Full-Stack E-Commerce Backend | [GitHub](#)

Node.js | Express | MongoDB | REST APIs

- Implemented backend services for authentication, product management, and order processing.
- Designed REST API endpoints to support secure transactions and database operations.
- Performed API testing with Postman and developed structured test cases to validate functionality.
- Implemented schema validation and error handling to improve reliability and data integrity.
- Used Git for version control and collaborative development.

Defect Detection Machine Learning Model

Python | Computer Vision | Microsoft Azure

- Developed a supervised machine learning model to classify manufacturing defects from labeled image datasets.
- Performed dataset preprocessing, feature extraction, and model evaluation using performance metrics.
- Deployed experiments on Microsoft Azure to support scalable training and testing workflows.

Clinic Management System

C++ | Object-Oriented Programming

- Developed a C++ application to manage clinic operations, including doctor scheduling, room assignments, and patient workflow coordination.
- Implemented object-oriented design with classes for doctors, rooms, and scheduling logic.
- Designed algorithms to assign doctors to available rooms and shifts while preventing scheduling conflicts.
- Applied structured debugging and testing to ensure accurate scheduling and system reliability.

Streaming Platform Web Application (Netflix-Style UI) | [GitHub](#)

React | JavaScript | REST APIs

- Developed a responsive streaming interface using React component architecture and dynamic routing.
- Integrated third-party REST APIs to fetch and display movie data in real time.
- Implemented reusable UI components and optimized rendering performance for smooth user interaction.
- Managed application state and asynchronous API requests using modern JavaScript patterns.
- Deployed the application and managed version control using Git/GitHub.

Apple-Inspired Landing Page

HTML | CSS | JavaScript | REST APIs | Render | GitHub

- Developed a responsive landing page inspired by Apple's product website using HTML, CSS, and JavaScript.
- Integrated YouTube API endpoints to dynamically fetch and display product videos on the page.
- Implemented modular JavaScript functions to handle asynchronous API requests and video rendering.
- Optimized layout responsiveness and media loading for a better experience across devices.

Portfolio & Web Solutions Platform

HTML | CSS | JavaScript | Hosting

- Developed and deployed responsive web applications and landing pages.
- Implemented modern UI components and optimized frontend performance.
- Managed domain configuration, hosting environments, and deployment using Hostinger and Bluehost.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, C, C++, SQL

Frameworks & Web Development: React, Node.js, Express, FastAPI, Flask, HTML, CSS, REST APIs

Cloud & DevOps: AWS EC2, Microsoft Azure, Google Cloud, Docker, containerization, Render, Hostinger, Bluehost

Databases: MongoDB, MySQL, Oracle SQL, SQLite, phpMyAdmin

AI & Data Systems: Machine Learning, Retrieval-Augmented Generation (RAG), LangChain, LLM APIs, NLP, Computer Vision

Software Engineering: Object-Oriented Programming, Data Structures & Algorithms, SDLC, Agile development, debugging, unit testing, integration testing, API testing (Postman), version control (Git), technical documentation, team collaboration

Systems & Networking: Linux/UNIX, operating system fundamentals, process memory concepts

Design & UX: Figma, user stories, user flows, accessibility-focused UI design